

Cardinality

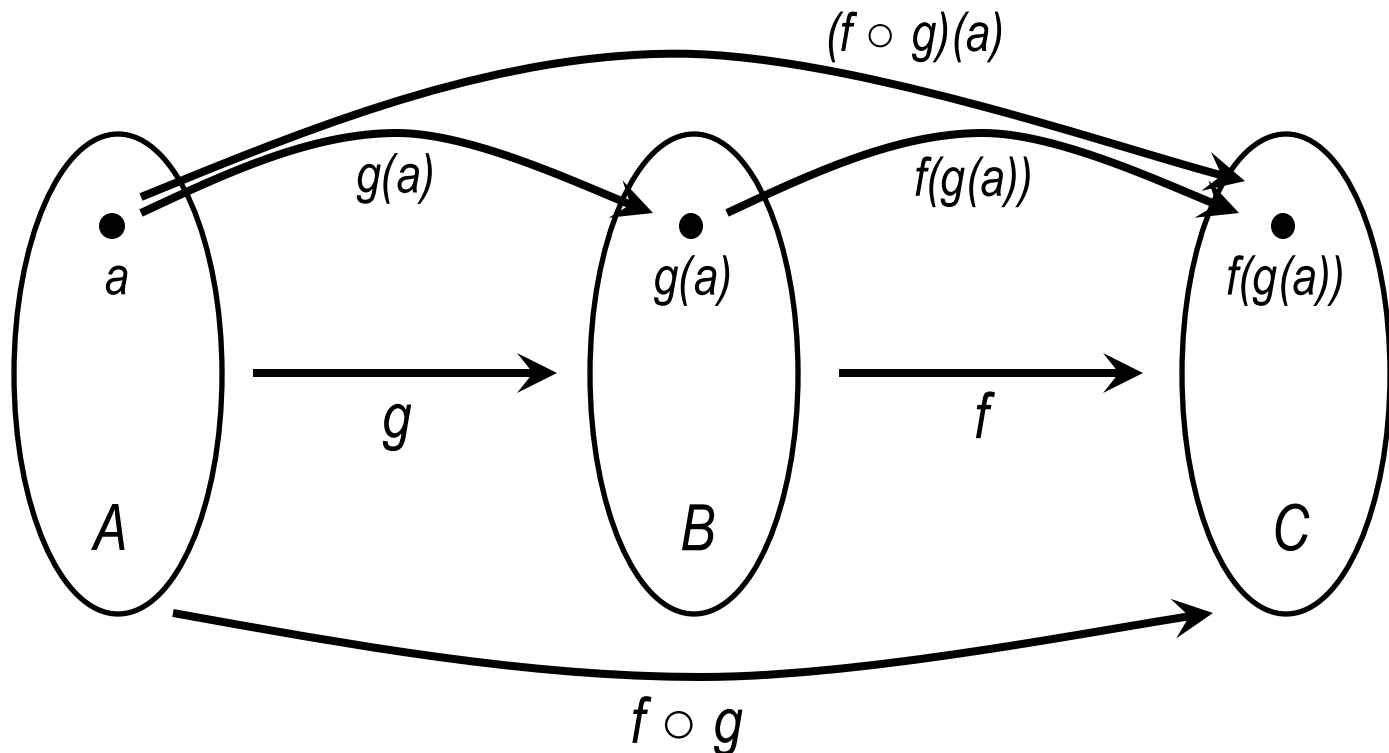
Properties of Functions

- A function f is said to be **one-to-one**, or **injective**, if and only if $f(a) = f(b)$ implies $a = b$.
- A function f from A to B is called **onto**, or **surjective**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function is called a **surjection** if it is onto.
- A function f is a **one-to-one correspondence**, or a **bijection**, if it is both one-to-one and onto.

Composition of Functions

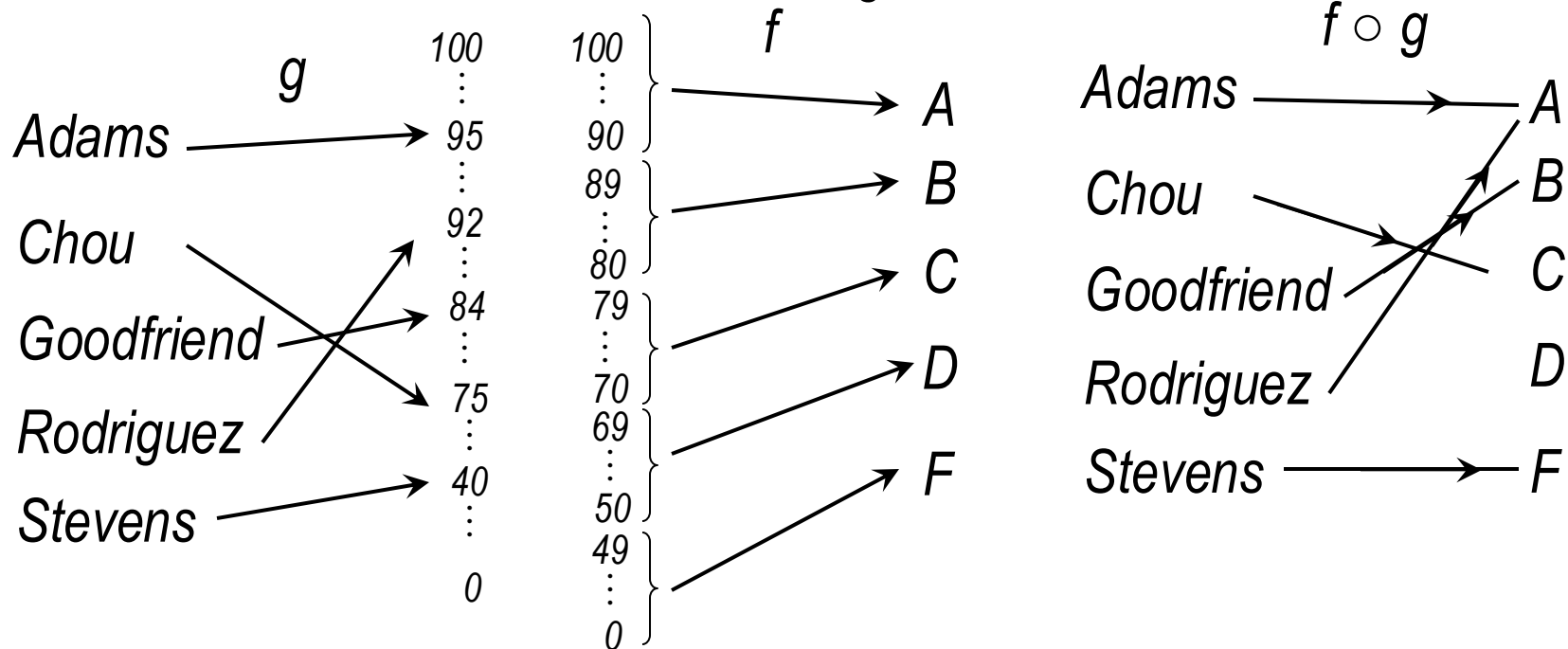
- Let g be a function from A to B and let f be a function from B to C . The **composition** of the functions f and g , denoted by $f \circ g$, is the function from A to C defined by

$$(f \circ g)(a) = f(g(a))$$



Composition of Functions (cntd)

- Suppose that the students first get numerical grades from 0 to 100 that are later converted into letter grade.



- Let $f(a) = b$ mean ' b is the father of a '.
What is $f \circ f$?

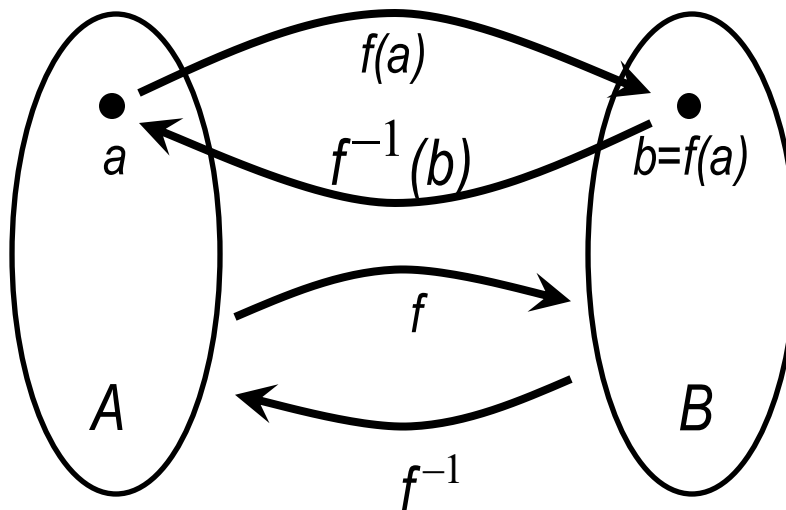
Composition of Numerical Functions

● Let $g(x) = x^2$ and $f(x) = x + 1$. Then
 $(f \circ g)(x) = f(g(x)) = g(x) + 1 = x^2 + 1$

● Thus, to find the composition of numerical functions f and g given by formulas we have to substitute x with $g(x)$ in $f(x)$.

Inverse Functions

- Let f be a bijection from the set A to the set B .
- The **inverse** of f , denoted f^{-1} , is the function that assigns to an element $b \in B$ the unique preimage $a \in A$ such that $f(a) = b$.
- Logically, $f^{-1}(b) = a$ if and only if $f(a) = b$



Note!
 f^{-1} does not mean $\frac{1}{f(x)}$

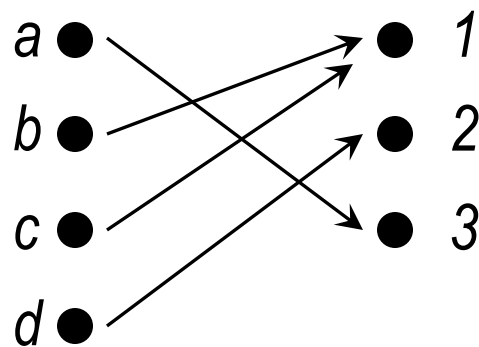
$$f \circ f^{-1} = i_B$$

$$f^{-1} \circ f = i_A$$

Inverse Functions (cntd)

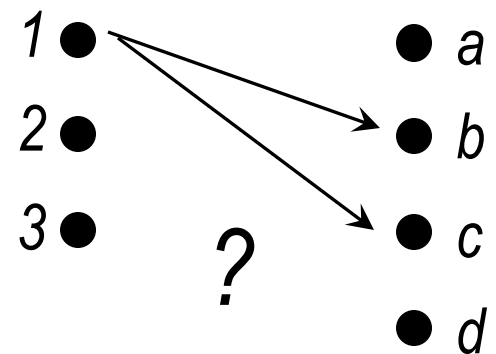
- If a function f is not a bijection, the inverse function does not exist.
Why?

- If f is not a bijection, it is either not one-to-one, or not onto



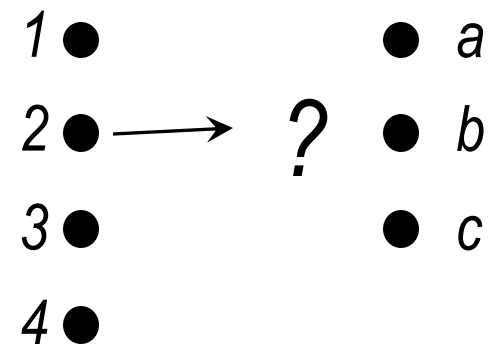
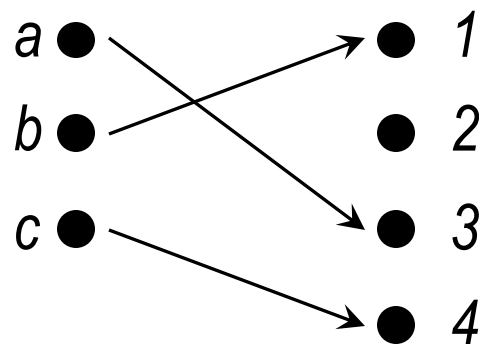
onto, not one-to-one

$$f^{-1}(1) = ?$$



one-to-one, not onto

$$f^{-1}(2) = ?$$

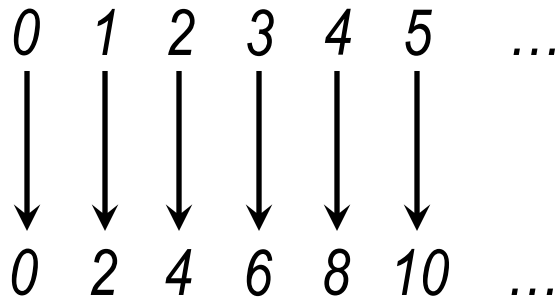


How to Count Elements in a Set

- *How many elements are in a set?*
- *Easy for finite sets, just count the elements.*
- *What about infinite sets? Does it make sense at all to ask about the number of elements in an infinite set?*
- *Can we say that this infinite set is larger than that infinite set?*
- *Which set is larger:*
 - *the set of all integers or the set of even integers?*
 - *the set of all integers or the set of all of all rationals?*
 - *the set of all integers or the set of all reals?*

Cardinality and Bijections

- Sets A and B (finite or infinite) have the same cardinality if and only if there is a bijection from A to B
- $|\mathbb{N}| = |2\mathbb{N}|$

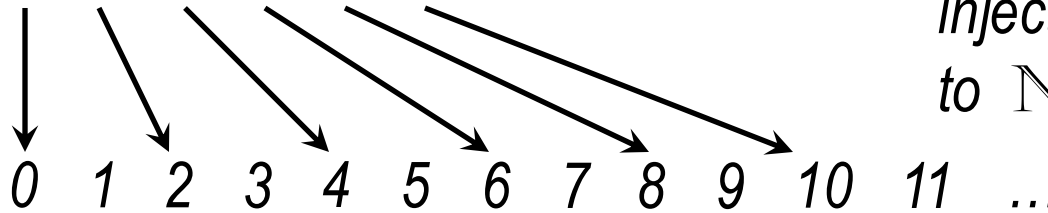


The function $f: \mathbb{N} \rightarrow 2\mathbb{N}$, where
 $f(x) = 2x$,
 is a bijection

Comparing Cardinalities

- Let A and B be sets. We say that $|A| \leq |B|$ if there is an injective function from A to B .

0 2 4 6 8 10 ...



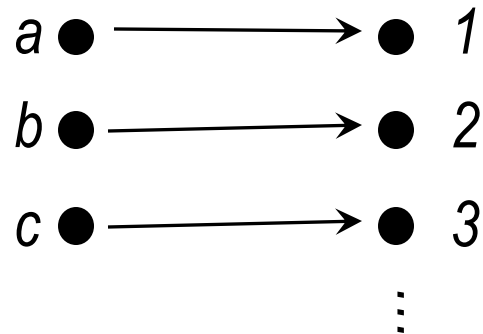
The function $f(x) = x$ is an injective function from $2\mathbb{N}$ to $\mathbb{N}$. Therefore $|2\mathbb{N}| \leq |\mathbb{N}|$

- If there is an injective function from A to B , but not from B to A , we say that $|A| < |B|$
- If there is an injective function from A to B and an injective function from B to A , then we say that A and B have the same cardinality
- Exercise: Prove that a bijection from A to B exists if and only if there are injective functions from A to B and from B to A .

Countable and Uncountable

- A set A is said to be **countable** if $|A| \leq |\mathbb{N}|$
- This is because an injective function from A to $\mathbb{N}$ can be viewed as assigning numbers to the elements of A , thus counting them
- Sets that are not countable are called **uncountable**
- Countable sets:

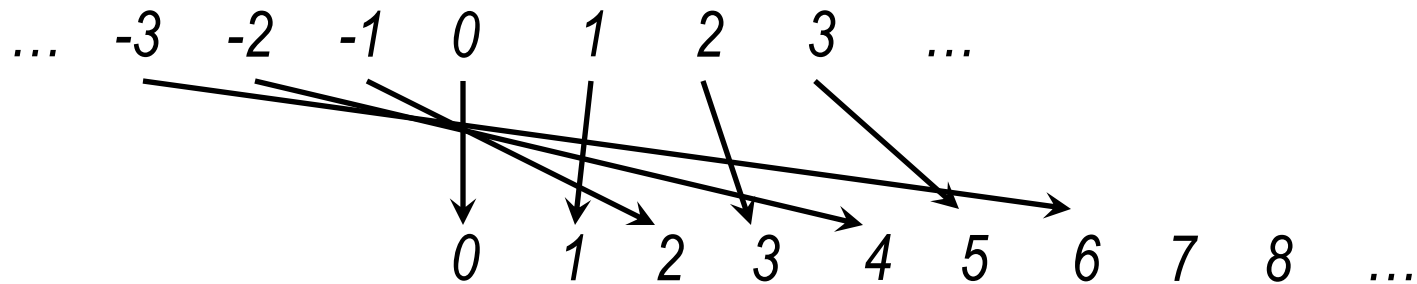
finite sets



any subset of $\mathbb{N}$

More Countable Sets

- *The set of all integers is countable*



- *In other words we can make a list of all integers*

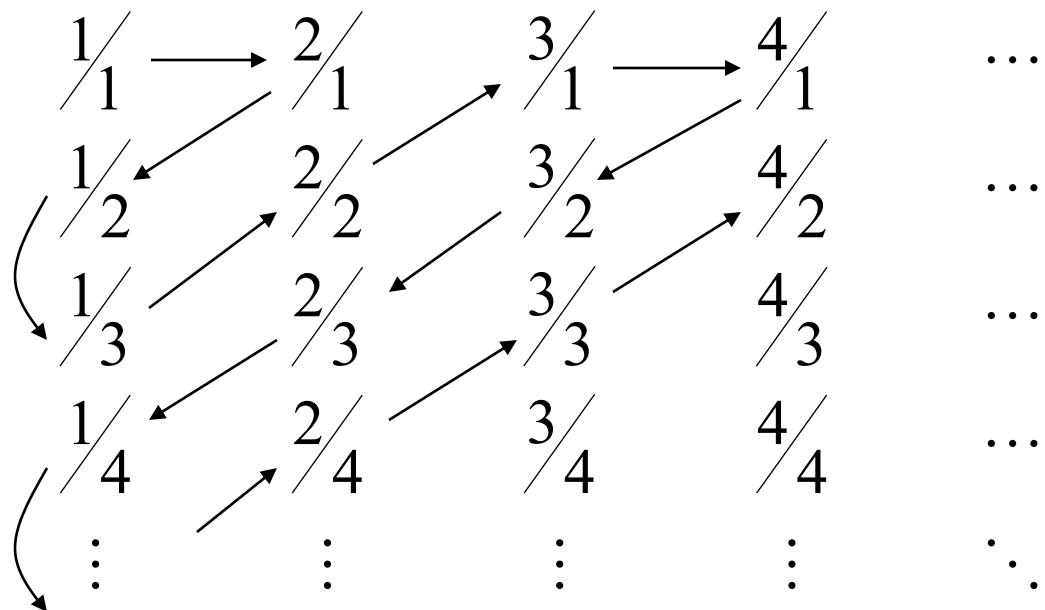
0, 1, -1, 2, -2, 3, -3, 4, -4, 5, -5, ...

- *The cardinality of the set of all natural numbers is denoted by $\aleph_0$*

"aleph-zero"

More Countable Sets (cntd)

- The set of positive rational numbers is countable
- Every rational number can be represented as a fraction $\frac{p}{q}$
We do not insist that p and q do not have a common divisor



- This gives an injection from $\mathbb{Q}^+$ to $\mathbb{N}$. The converse injection is $f(x) = x + 1$

The Smallest Infinite Set

● *Theorem.*

If A is an infinite set, then $|A| \geq \aleph_0$

● *Proof requires mathematical induction. Wait for a few days.*

Uncountable Sets

- *Can we make a list of all real numbers?*
- *Every real number can be represented as an infinite decimal fraction, like 3.1415926535897932384626433832795028841971...*
- *Suppose we have constructed a list of all real numbers*

1. $a_{10}.a_{11}a_{12}a_{13}a_{14}a_{15}a_{16}a_{17} \dots$
2. $a_{20}.a_{21}a_{22}a_{23}a_{24}a_{25}a_{26}a_{27} \dots$
3. $a_{30}.a_{31}a_{32}a_{33}a_{34}a_{35}a_{36}a_{37} \dots$
4. $a_{40}.a_{41}a_{42}a_{43}a_{44}a_{45}a_{46}a_{47} \dots$
5. $a_{50}.a_{51}a_{52}a_{53}a_{54}a_{55}a_{56}a_{57} \dots$
- $\vdots$

Here the a_{ij} are digits $0, 1, 2, \dots, 9$

Let

$$b_i = \begin{cases} 4, & \text{if } a_{ii} \neq 4, \\ 5 & \text{otherwise} \end{cases}$$

- *It is not hard to see that the number $0.b_1b_2b_3b_4b_5b_6b_7 \dots$ is not in this list*

Cantor's Theorem

● **Theorem (Cantor).** *For any set $|P(A)| > |A|$.*

Proof.

It is easy to see that $|P(A)| \geq |A|$. (Find an injection $A \rightarrow P(A)$)

Prove $|P(A)| \neq |A|$. Suppose there is a bijection $f: A \rightarrow P(A)$.

We find a set that does not belong to the range of f . A contradiction with the assumption that f is bijective.

Consider the set $T = \{a \in A \mid a \notin f(a)\}$

If T is in the range of f , then there is $t \in A$ such that $f(t) = T$.

Either $t \in T$ or $t \notin T$.

If $t \in T$ then $t \in f(t)$, and we get $t \notin T$.

If $t \notin T$ then $t \in T$.

Q.E.D.

Cantor's Theorem (cntd)

- *This method is called Cantor's diagonalization method*
- *The cardinality of $P(A)$ is denoted by $2^{|A|}$*
- *Thus, we obtain an infinite series of infinite cardinals*

$$|\mathbb{N}| = \aleph_0$$

$$2^{\aleph_0} = \aleph_1 \quad (= |\mathbb{R}|)$$

$$2^{\aleph_1} = \aleph_2$$

$$\vdots$$

Continuum Hypothesis

- *We just proved that $\aleph_0 < |\mathbb{R}|$. Does there exist a set A such that $\aleph_0 < |A| < |\mathbb{R}|$?*
- *The negative answer to this question is known as the **continuum hypothesis**.*
- *Continuum hypothesis is the first problem in the list of Hilbert's problems*
- *Paul Cohen resolved the question in 1963.
...It can't be proven or disproven in ZFC.*



Practice

- *Determine whether each of the following statements is true:*
 - *If A, B are countable sets then $A \cup B$ is countable*
 - *If A, B are uncountable sets then $A \cap B$ is uncountable*
 - *If A, B are countable sets then $A - B$ is countable*
- *Find a one-to-one correspondence between $\mathbb{Z}^+$ and $\{2, 6, 10, 14, \dots\}$*
- *Let $I = \{r \in \mathbb{R} \mid r \text{ is irrational}\} = \mathbb{R} - \mathbb{Q}$. Is I countable or uncountable?*
- *Construct a bijective mapping between the closed interval $[0; 1]$ and the square $[0; 1] \times [0; 1]$*